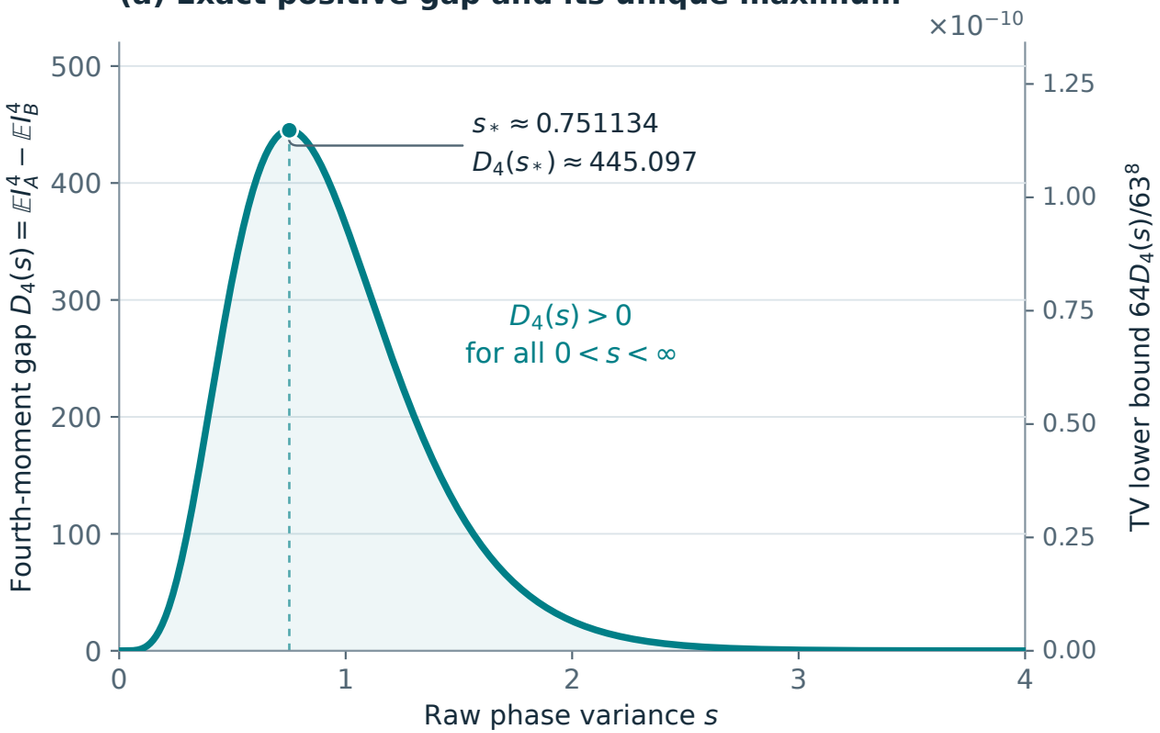
Higher-order discrimination under identical Gaussian site noise

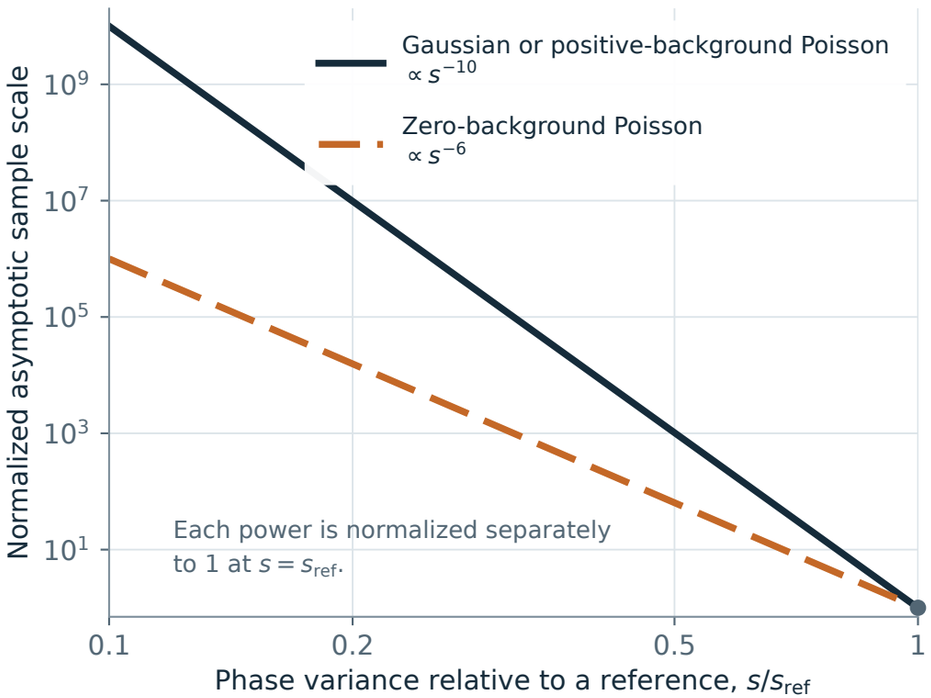
63-site homometric pair at the common amplitude null $q = \pi/3$; s is the raw Gaussian phase variance.

The first three intensity moments coincide. The fourth moment separates the laws for every finite $s > 0$.

(a) Exact positive gap and its unique maximum



(b) Sharp small-noise sample orders



$$D_4(s) = \frac{455015761}{20000}\, e^{-4s}(1 - e^{-s})^5(e^{-3s} + 3e^{-2s} + 6e^{-s} + 6)$$

Both endpoint limits are zero: $D_4(s) \sim \frac{455015761}{1250} s^5$ as $s \downarrow 0$; $D_4(s) \sim \frac{1365047283}{10000} e^{-4s}$ as $s \to \infty$.

Panel (a): the right axis is a conservative lower bound, not the actual total variation. The maximum optimizes this moment witness.
Panel (b): independent observations, fixed calibrated detector parameters, and fixed success above chance. Asymptotic powers, not finite sample forecasts.